

HR Employee Attrition Analysis Report

Excel Dashboard and Business Insights

1. Executive Summary

This project analyzes employee attrition using an HR employee dataset of 1,470 records. The objective is to identify which employee groups have higher attrition and what factors may be associated with employees leaving the organization. The analysis was completed in Microsoft Excel using pivot tables, calculated attrition rates, KPI cards, and dashboard charts.

The overall attrition rate is 16.12%. The strongest attrition signals were found among employees working overtime, employees under 30, single employees, and Sales Representative job roles.

2. Key Performance Indicators

Metric	Value	Business Meaning
Total Employees	1,470	Total employee records analyzed
Employees Left	237	Employees with Attrition = Yes
Active Employees	1,233	Employees with Attrition = No
Overall Attrition Rate	16.12%	Percentage of employees who left

3. Dataset Overview

Item	Description
Dataset	HR Employee Attrition dataset
Rows	1,470 employee records
Columns	35 HR-related fields
Important Fields	Age, Attrition, Department, Gender, JobRole, MonthlyIncome, OverTime, YearsAtCompany, JobSatisfaction, WorkLifeBalance
Tool Used	Microsoft Excel

4. Methodology

- Created pivot tables to summarize attrition by employee groups.
- Calculated attrition rate using: $\text{Employees Left} / \text{Total Employees}$.
- Compared attrition rates instead of only attrition counts, because departments and job roles have different employee sizes.
- Built dashboard visuals for Department, Job Role, OverTime, Age Group, Gender, and Marital Status.

5. Detailed Findings

5.1 Attrition by Department

Department	Stayed	Left	Total Employees	Attrition Rate
Human Resources	51	12	63	19.05%
Research & Development	828	133	961	13.84%
Sales	354	92	446	20.63%

Insight: Sales has the highest attrition rate at 20.63%. Research & Development has the highest number of employees leaving, but Sales has the highest percentage of employees leaving.

5.2 Attrition by Job Role

Job Role	Left	Total Employees	Attrition Rate
Sales Representative	33	83	39.76%
Laboratory Technician	62	259	23.94%
Human Resources	12	52	23.08%
Sales Executive	57	326	17.48%
Research Scientist	47	292	16.10%
Manufacturing Director	10	145	6.90%
Healthcare Representative	9	131	6.87%
Manager	5	102	4.90%
Research Director	2	80	2.50%

Insight: Sales Representative has the highest attrition rate at 39.76%, which is a serious retention concern.

5.3 Attrition by OverTime

OverTime	Stayed	Left	Total Employees	Attrition Rate
No	944	110	1,054	10.44%
Yes	289	127	416	30.53%

Insight: Employees working overtime have an attrition rate of 30.53%, almost three times higher than employees not working overtime.

5.4 Attrition by Age Group

Age Group	Stayed	Left	Total Employees	Attrition Rate
Under 30	235	91	326	27.91%
30-40	585	94	679	13.84%
41-50	288	34	322	10.56%
50+	125	18	143	12.59%

Insight: Employees under 30 have the highest attrition rate at 27.91%, suggesting younger employees may be more likely to leave.

5.5 Attrition by Gender

Gender	Stayed	Left	Total Employees	Attrition Rate
Female	501	87	588	14.80%
Male	732	150	882	17.01%

Insight: Male employees show a slightly higher attrition rate than female employees, but the gap is not large compared with OverTime or Job Role.

5.6 Attrition by Marital Status

Marital Status	Stayed	Left	Total Employees	Attrition Rate
Divorced	294	33	327	10.09%
Married	589	84	673	12.48%
Single	350	120	470	25.53%

Insight: Single employees have the highest attrition rate at 25.53%, more than double the attrition rate of divorced employees.

6. Business Recommendations

- **Reduce overtime pressure:** OverTime employees have a much higher attrition rate. Review workloads, staffing levels, and overtime policies.
- **Focus on Sales Representative retention:** Sales Representatives have the highest job-role attrition rate. Investigate targets, incentives, onboarding quality, and career growth opportunities.

- **Support younger employees:** Employees under 30 show high attrition. Improve mentoring, career path clarity, training, and early-stage employee engagement.
- **Review Sales department retention strategy:** Sales has the highest department-level attrition rate. Conduct exit interviews and manager-level reviews to identify specific causes.
- **Monitor single employees as a risk group:** Single employees show higher attrition. Review engagement patterns and flexibility needs without making assumptions at an individual level.

7. Conclusion

The HR attrition analysis shows that attrition is not evenly distributed across the workforce. OverTime, Job Role, Age Group, Department, and Marital Status show clear differences in attrition rates. The most important business finding is that overtime employees and Sales Representatives are leaving at much higher rates. These areas should be prioritized for deeper investigation and retention planning.

8. Resume Project Description

HR Employee Attrition Analysis | Excel

Analyzed 1,470 employee records using Excel Pivot Tables and dashboard charts to identify attrition patterns across departments, job roles, overtime status, age groups, gender, and marital status. Calculated attrition rates and generated business recommendations to support employee retention strategy.